

GROUPS OF MATRICES

(MA1) Is the group $GL_2(\mathbb{Z}[t, t^{-1}])$, i.e., the group of invertible matrices over the ring of one-variable Laurent polynomials with integral coefficients, generated by elementary and diagonal matrices?

*(MA2) Find a particular matrix from $GL_2(\mathbb{Z}[t, t^{-1}, s, s^{-1}])$, which is not a product of elementary and diagonal matrices. [Background](#)

(MA3) (a) The subgroup membership problem for the group $SL_3(\mathbb{Z})$. [Background](#)

(b) The subgroup membership problem for the group $SL_2(\mathbb{Q})$.

(MA4) (S.Thomas) Does there exist a simple torsion-free linear group?

(MA5) (A.L.Shmelkin) Is it true that identities of any linear group have a finite basis?

(MA6) (Yu. Merzlyakov) Is there a rational number a , $|a| < 2$, such that two 2×2 matrices $\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}$ generate a free group? [Background](#)

(MA7) (L. Vaserstein) For a ring R , let $E_n(R)$ be the subgroup of $GL_n(R)$ generated by elementary matrices. Is there a ring R and an element of $E_n(R)$ (for some n) which is not a commutator?